

■■ INDIA CPI INFLATION ANALYSIS

Understanding Inflation Trends, COVID-19 Impact &
Crude Oil Price Relationships

Data Analytics Case Study

Project Focus	India Consumer Price Index (CPI)
Analysis Period	2013–2023 historical CPI dataset; focused analyses through May 2023
Primary Tool	Microsoft Excel
Techniques	Data Cleaning, Trend Analysis, MoM/YoY Analysis, Correlation Analysis, Visualization
Portfolio Purpose	Demonstrate practical data analysis and insight generation

Prepared as a portfolio-ready documentation report based on the accompanying Excel workbook.

1. Executive Summary

This case study analyzes India's Consumer Price Index (CPI) using historical category-level CPI data. The project examines the composition of the CPI basket, annual inflation trends, food inflation behavior, the change in inflation around the onset of COVID-19, and the relationship between imported crude oil price movements and transport inflation.

The analysis indicates that **Food & Beverages** is the largest broad contributor in the analyzed CPI basket at approximately **34%**. Annual inflation is highest in the workbook's selected yearly series in **2022 (6.8%)**, while the 12-month food-inflation series reaches its highest value in **November 2022 (7.2%)**.

The COVID-19 comparison shows inflation increasing from **4.0%** before March 2020 to **6.1%** after March 2020. The crude-oil analysis reports a very strong positive correlation of approximately **0.989** between the selected oil-price-change and transport-inflation series, indicating a strong relationship within the analyzed observations.

These results demonstrate how spreadsheet-based analysis can transform raw economic data into structured findings that support economic interpretation and decision-making.

Note: The report documents the results contained in the submitted workbook. Correlation describes association in the analyzed data and should not by itself be interpreted as proof of causation.

2. Dataset & Methodology

The workbook contains **372 data records** plus headers in both the raw and cleaned datasets, covering Rural, Urban, and Rural+Urban sectors. The dataset includes monthly CPI index values across food, clothing, housing, fuel, health, transport and communication, education, personal care, and other categories.

The workbook is organized into dedicated sheets for the introduction, raw data, cleaned data, cleaning process, five analytical problems, and the final conclusion. This structure provides a clear audit trail from source data through analysis and findings.

Data preparation steps documented in the workbook include:

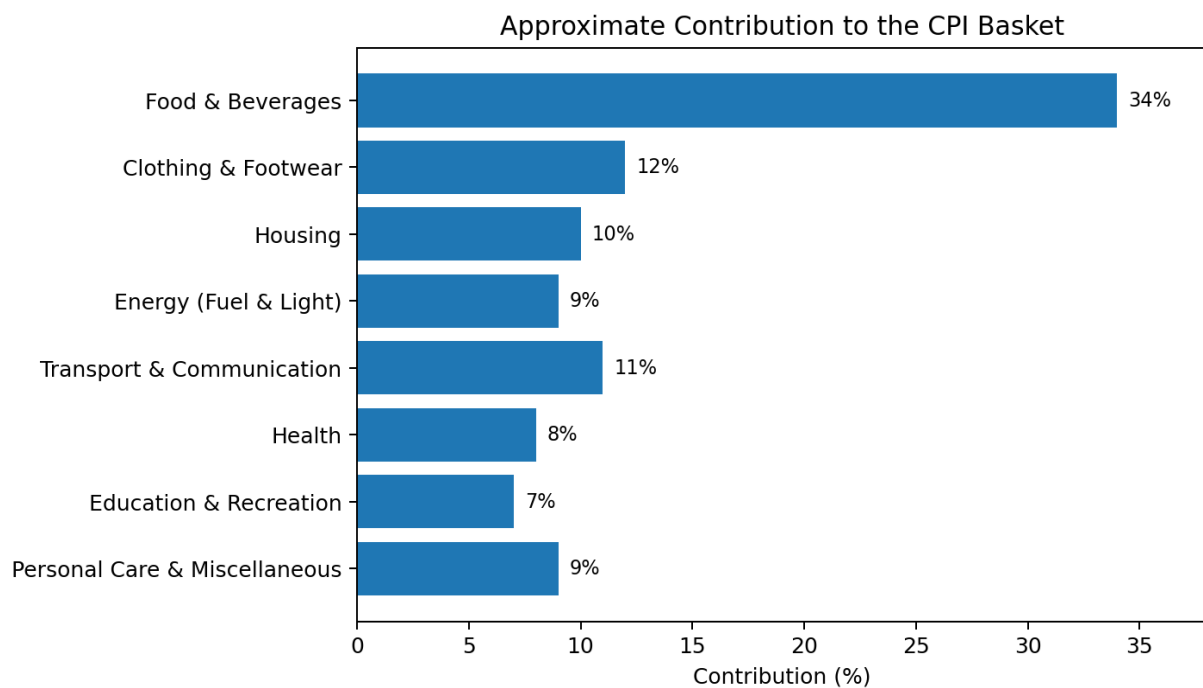
Step	Completed Activity
1	Imported the CPI dataset into Excel
2	Checked for missing/null values
3	Standardized month and year formatting
4	Removed unnecessary blank rows and duplicate records
5	Validated numerical CPI formatting
6	Organized data into category-wise analytical structure

Analytical methods used

Analysis	Method
CPI basket contribution	Category value relative to total CPI value
Annual inflation	Year-on-year comparison of Rural + Urban general CPI
Food inflation	Monthly percentage-change analysis over the 12-month period ending May 2023
COVID-19 impact	Comparison of inflation before and after March 2020
Crude oil relationship	Excel CORREL() correlation between oil-price changes and CPI category inflation

3. CPI Basket Contribution Analysis

The first analysis identifies the approximate contribution of broader categories to the CPI basket using the latest available CPI values represented in the workbook.

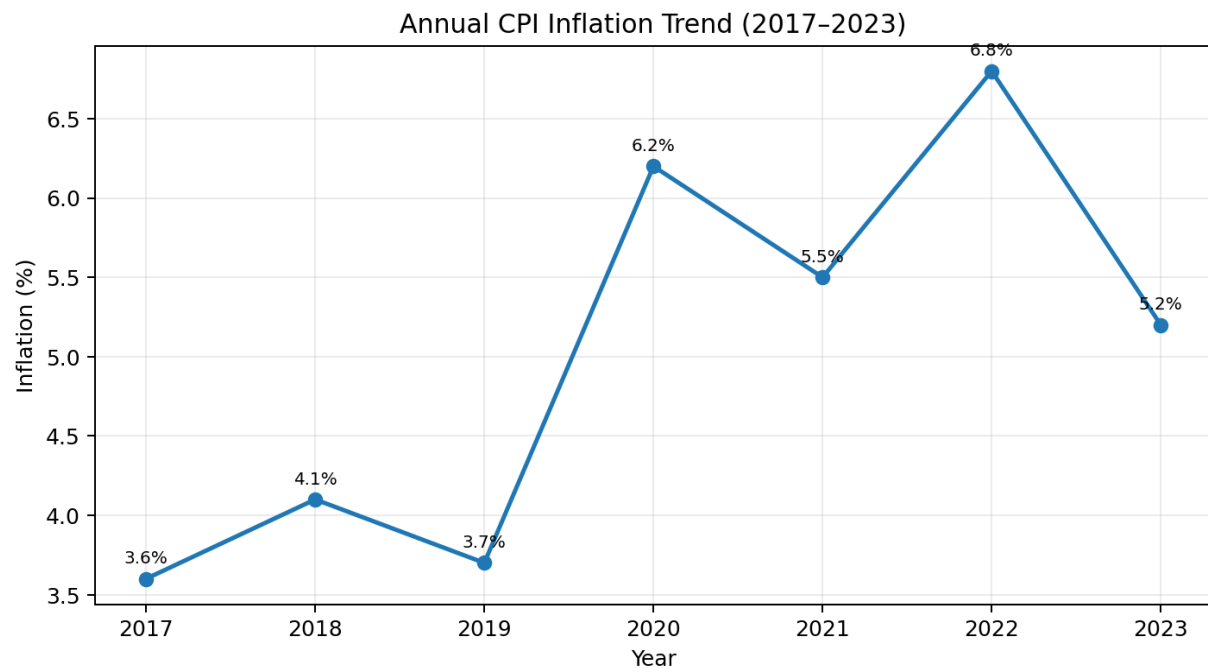


Broad Category	Approx. Contribution %
Food & Beverages	34%
Clothing & Footwear	12%
Housing	10%
Energy (Fuel & Light)	9%
Transport & Communication	11%
Health	8%
Education & Recreation	7%
Personal Care & Miscellaneous	9%

Key insight: Food & Beverages has the largest approximate contribution at 34%, making it the most prominent broad category in this analysis. Clothing & Footwear follows at 12%, while Transport & Communication contributes approximately 11%.

4. Annual CPI Inflation Trend

The second analysis evaluates the year-on-year inflation trend for the combined Rural + Urban sector beginning in 2017. The workbook reports the following yearly inflation values:

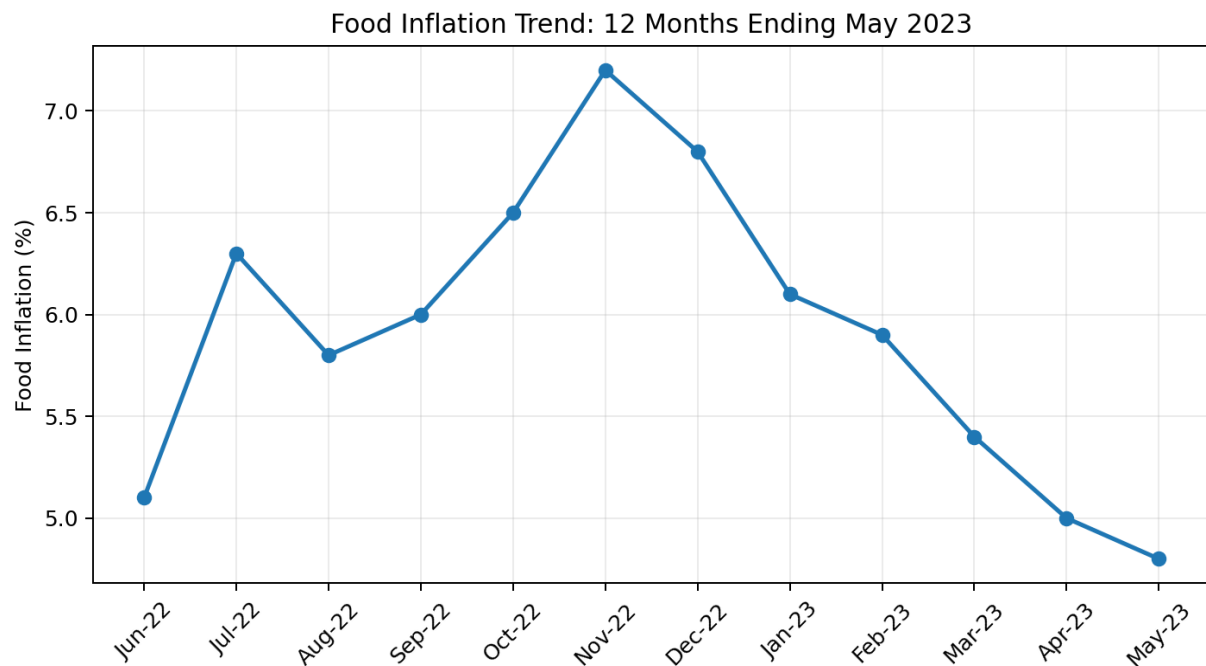


Year	Inflation Trend %
2017	3.6%
2018	4.1%
2019	3.7%
2020	6.2%
2021	5.5%
2022	6.8%
2023	5.2%

Key findings: Inflation rises from 3.6% in 2017 to 4.1% in 2018, then reaches 6.2% in 2020. The highest value in the analyzed yearly series is **6.8% in 2022**, followed by 5.2% in 2023.

5. Food Inflation Analysis

The third analysis studies food inflation over the 12-month period ending in May 2023. Monthly changes were examined to understand the behavior of food inflation and identify the highest and lowest observations.

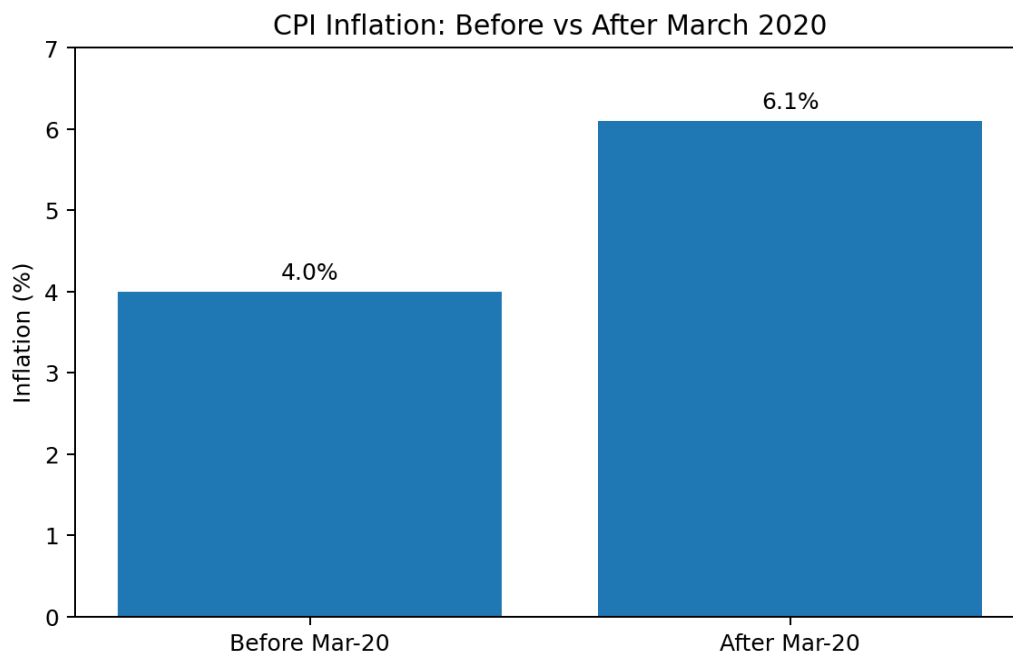


Month	Food Inflation %
Jun-22	5.1%
Jul-22	6.3%
Aug-22	5.8%
Sep-22	6.0%
Oct-22	6.5%
Nov-22	7.2%
Dec-22	6.8%
Jan-23	6.1%
Feb-23	5.9%
Mar-23	5.4%
Apr-23	5.0%
May-23	4.8%

Key findings: Food inflation reaches its highest reported value of **7.2% in November 2022**. It then trends downward overall, reaching the lowest reported value of **4.8% in May 2023**. This indicates a clear easing in the food-inflation series over the latter part of the period.

6. COVID-19 Impact on Inflation

The fourth analysis investigates how inflation changed around the onset of the COVID-19 pandemic, using March 2020 as the reference point and focusing on the broader effect on inflation and essential categories.



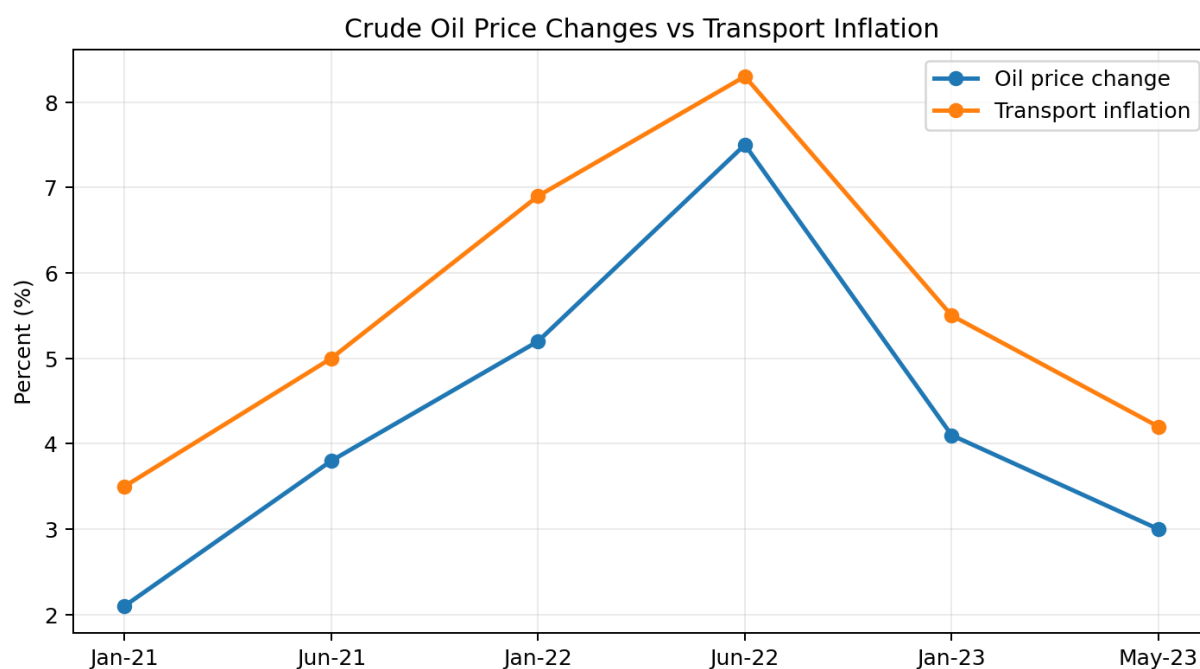
Period	Inflation %
Before Mar-20	4.0%
After Mar-20	6.1%

Change observed: The reported inflation measure increases by **2.1 percentage points**, from 4.0% before March 2020 to 6.1% after March 2020. The workbook associates this period with supply-chain disruptions, higher healthcare demand, and rising prices of essential goods.

The comparison is descriptive: it highlights a change around the pandemic onset, but it does not isolate COVID-19 as the only possible driver of inflation.

7. Crude Oil Prices & Transport Inflation

The fifth analysis examines the relationship between imported crude oil price movements and CPI category inflation between 2021 and 2023. The workbook uses Excel's **CORREL()** function to quantify the relationship.



Month	Oil Price Change %	Transport Inflation %
Jan-21	2.1%	3.5%
Jun-21	3.8%	5.0%
Jan-22	5.2%	6.9%
Jun-22	7.5%	8.3%
Jan-23	4.1%	5.5%
May-23	3.0%	4.2%

Correlation result: The workbook reports a correlation coefficient of approximately **0.989**. A value this close to +1 indicates a very strong positive association in the analyzed observations: higher oil-price changes are accompanied by higher transport inflation values in the selected series.

Correlation should be interpreted as association rather than causation. A broader study would ideally use a larger, consistently timed monthly dataset and control for other factors affecting transport and inflation.

8. Overall Findings & Conclusion

The India CPI Inflation case study demonstrates how category-level economic data can be analyzed to understand the major components and drivers associated with inflation.

Area	Main Finding
CPI basket	Food & Beverages is the largest broad contribution at approximately 34%.
Annual inflation	The highest value in the 2017–2023 series is 6.8% in 2022.
Food inflation	The series peaks at 7.2% in Nov-22 and falls to 4.8% by May-23.
COVID-19 period	Inflation increases from 4.0% before Mar-20 to 6.1% after Mar-20.
Crude oil relationship	Oil-price changes and transport inflation show a reported correlation of about 0.989.

Overall, the project finds that **food prices play a major role in India's CPI basket**, while inflation behavior can also be influenced by major disruptions and global commodity movements. The strong oil–transport association in the selected observations reinforces the importance of monitoring energy and transportation-related price pressures.

From a data analytics perspective, the project demonstrates practical skills in **Excel-based data cleaning, structured analysis, percentage calculations, trend analysis, correlation analysis, and data visualization**. The workbook and this report together provide a reproducible portfolio artifact showing the journey from raw CPI data to interpretable insights.

Future scope

Future analysis could extend the study with a larger monthly crude-oil dataset, statistical significance testing, lagged correlations, regression analysis, seasonal decomposition, and interactive Power BI dashboards. These additions could help distinguish correlation from causation and provide a more granular view of inflation drivers.

End of Report